

Assignment:- The five numbers summary for the day and night classes is

	Minimum	Q_1	Median	Q_3	Maximum
Day	32	56	74.5	82.5	99
Night	25.5	78	81	89	98

Sol:-

Day $Q_1 - Q_3 = IQR$

$$IQR = Q_3 - Q_1$$

$$= 82.5 - 56$$

$$= 26.5$$

$$IQR * 1.5 = 39.75$$

$$\text{Lower outlier} = Q_1 - (1.5 * IQR)$$

$$= 56 - 39.75$$

$$= 16.25$$

$$\text{Greater outlier} = Q_3 + (1.5 * IQR)$$

$$= 82.5 + 39.75$$

$$= 122.25$$

Since, the
of day and
maximum

Night:-

It

It

Lower

Greater

greater for
one but
Some

61

Since, the lesser outlier is less than the minimum of day and greater outlier is greater than the maximum, there is no outliers present in day.

Night:-

$$\begin{aligned} IQR &= Q_3 - Q_1 \\ &= 89 - 78 \\ &= 11 \end{aligned}$$

$$IQR * 1.5 = 16.5$$

$$\begin{aligned} \text{Lesser outlier} &= Q_1 - (1.5 * IQR) \\ &= 78 - 16.5 \\ &= 61.5 \end{aligned}$$

$$\begin{aligned} \text{Greater outlier} &= Q_3 + (1.5 * IQR) \\ &= 89 + 16.5 \\ &= 105.5 \end{aligned}$$

Here in night, we do not have greater ~~lesser~~ outlier since it is lesser than actual one but we have the lesser outlier (ie) 25.5. So we must re order it as follows:-

61.5, 78, 81, 89, 98.